

IAL Biology Unit 4 (WBI14): Rapid Revision

Energy, Environment, Microbiology and Immunity · last-minute summary · 1 h 45 min · 90 marks · 8–9 questions · 1–3 asterisked (*) levels-marked questions · A* needs grade A overall *and* 90% across Units 4, 5 and 6

WHERE THE MARKS ARE

Photosynthesis (12–20) · immunity (10–18) · microbiology (10–18) · climate change (8–16) · energy flow (6–14). **These five appear every paper**; add evolution and forensics and you have 70–80 of the 90 marks.

WHAT SEPARATES THE GRADES

1. Use the examiner's word, not the everyday one. **2.** Link points with *because, so, therefore* — six unlinked facts score four. **3.** Quote **figures with units**. **4.** On every (*) question, **pair each description with its explanation**; describing alone caps you at Level 1–2.

PHOTOSYNTHESIS

Light-dependent reactions occur in the **thylakoid membranes**; the Calvin cycle in the **stroma**.

THE LIGHT-DEPENDENT CHAIN

1. Light absorbed by chlorophyll in the photosystems. **2.** Electrons **excited and released**. **3.** Electrons pass along the **electron transport chain, releasing** energy. **4.** Energy **pumps protons into the thylakoid space**, creating a gradient. **5.** Protons diffuse back through **ATP synthase** — chemiosmosis — making ATP. **6.** NADP accepts electrons and protons → **reduced NADP**.

Photolysis: $2\text{H}_2\text{O} \rightarrow 4\text{H}^+ + 4\text{e}^- + \text{O}_2$ — replaces lost electrons, supplies protons, releases oxygen.

Learn verbatim (a repeated 2-mark answer): pigments absorb light “**so that the electrons will be excited and released from the photosystems**” and “**so that reduced NADP and ATP can be synthesised**”.

Calvin cycle	What happens
Carbon fixation	$\text{CO}_2 + \text{RuBP (5C)} \xrightarrow{\text{RUBISCO}} 2 \times \text{GP (3C)}$
Reduction	$\text{GP} + \text{reduced NADP} + \text{ATP} \rightarrow \text{GALP (3C)}$
Regeneration	5 of every 6 GALP + ATP regenerate RuBP; 1 in 6 leaves

Per hexose: **6 turns, 6 CO₂, 18 ATP, 12 reduced NADP**.

RATE CHANGES — THE FIVE-LINK CHAIN

More light → **more photolysis** → more electrons released → **more ATP and reduced NADP** → Calvin cycle faster, **RuBP regenerated faster to bind more CO₂**.

Reverse case (net CO₂ uptake negative): respiration exceeds photosynthesis; **RuBP is regenerated too slowly to bind as much CO₂**. **Do not accept “no photosynthesis”** — the RuBP clause is the discriminating mark.

Chloroplast: double membrane; thylakoids in grana carrying photosystems, pigments and ATP synthase (large surface area); a **small thylakoid space** so the proton gradient builds quickly; stroma with RUBISCO; **DNA loop** and **70S ribosomes**; starch grain.

Absorption spectrum = light absorbed per wavelength; **action spectrum** = rate of photosynthesis per wavelength. Similar shapes show that the pigments absorbing light drive photosynthesis.

Chromatography: R_f = distance moved by the pigment ÷ distance moved by the solvent front. Order: carotene (highest, most soluble in the solvent), xanthophyll, chlorophyll a, chlorophyll b.

Rate practical: *Elodea* or *Cabomba* in hydrogencarbonate solution; a **photosynthometer** beats counting bubbles; a **heat shield** so only light intensity changes; allow 5 minutes to equilibrate. The gas is not pure oxygen and some dissolves, so the volume **underestimates**.

ENERGY FLOW AND ECOLOGY

THE DEFINITIONS

GPP = total energy fixed by photosynthesis per unit area per unit time. **R** = energy used in respiration. **NPP** = **GPP** – **R**. Units **kJ m⁻² yr⁻¹** — they must contain an **area and a time**. In a table **GPP is always largest**.

Efficiency (%) = (energy in the higher level ÷ energy in the lower) × 100.

Why so little transfers: not all is eaten (bones, roots, bark); not all is digested — egested in faeces; heat lost in respiration, especially by endotherms; losses in excretion. So food chains rarely exceed four or five links, and eating plants is more efficient than eating meat.

Term	The accepted definition
Habitat	The place where an organism lives
Population	All the organisms of one species in a habitat at one time
Community	All the populations of different species living and interacting
Ecosystem	Organisms interacting with each other and with the abiotic environment — both halves needed; “habitat” is ignored
Niche	The role an organism plays, plus a stated reason when a context is given
Biodiversity	Number of species and genetic diversity within a species

Succession: pioneers colonise bare rock → die and decompose → **humus and thin soil** → soil holds water and nutrients → new species outcompete the pioneers → **biodiversity increases at each sere** → stable **climax community**. Secondary succession is faster because soil, nutrients and a seed bank already exist.

CLIMATE CHANGE

Evidence	Mechanism and limitation
CO ₂ records	Air bubbles in ice cores, dated by annual layers. Polar regions only; dating uncertain at depth
Temperature	Thermometers for ~150 years; before that ¹⁸ O: ¹⁶ O ratios — more ¹⁸ O means warmer. Short record; uneven stations
Pollen in peat bogs	Peat is acidic and anaerobic → slow decomposition → pollen preserved; species-specific shapes; deeper is older. Pollen may blow in; tolerances may have changed
Dendrochronology	Dating per year; wider means warmer or wetter; overlapping samples extend the record. Width also depends on nutrients, disease and competition

“ANTHROPOGENIC CLIMATE CHANGE” — 2 MARKS, 2 CLAUSES

“Caused by the **effects of humans**” — **named activities are ignored** — and “changes to **mean temperature or rainfall**” (long-term change in weather patterns); “**weather**” **unqualified is ignored**. Listing “burning fossil fuels” earns **nothing**.

The greenhouse effect, four steps: short-wavelength radiation passes through and is absorbed by the surface → the surface **re-radiates at longer wavelengths (infrared)** → greenhouse gases absorb infrared and re-radiate some back → less energy escapes, so mean surface temperature rises. Gases: CO₂, methane, water vapour, nitrous oxide, CFCs. **Oxygen and nitrogen are not greenhouse gases.**

Methane chain: microorganisms secrete enzymes → organic matter broken down → sugars respired → **anaerobic conditions produce methane** → methane traps infrared → temperature of **the atmosphere, surface or water rises** — “**global temperatures**” is ignored.

Effects on organisms, three categories: distribution (move polewards or to altitude; those at the poles or on mountain tops have nowhere to go); **development** (most are ectothermic, so development is faster up to the optimum); **life cycles** (earlier flowering, egg-laying and emergence, risking **desynchronisation** with the food supply).

Q_{10} = rate at $(T + 10^\circ\text{C}) \div$ rate at T — **no units** (“do not accept with units”), and only meaningful below the optimum.

Why a prediction is unreliable: the data fluctuate; factors increasing or decreasing warming could change; **the trend itself could change**; older data are less accurate.

EVOLUTION AND RESISTANCE

THE 3-MARK NATURAL SELECTION CHAIN, AND ITS TWO FATAL ERRORS

A **mutation** in the DNA produces an **allele for resistance**; the chemical acts as a **selection pressure**; **resistant** individuals survive and **pass the allele to their offspring**.

Do not accept “immune” for “resistant”. Do not accept that the selection pressure causes the mutation — mutation is random and comes **first**.

Resistance evolves fast in bacteria because of a short life cycle, a high mutation rate, many offspring and a strong selection pressure; they also spread it **horizontally by conjugation via plasmids**.

Evolution = a change in **allele frequency** in a population. **Allopatric** = a geographical barrier; **sympatric** = same area, ecological, behavioural or temporal isolation. Full chain: **reduced gene flow** → **changes in allele frequency** → **reproductive isolation** → **two species**. Read carefully which of the two chains is being asked for.

MICROBIOLOGY

“DESCRIBE HOW BACTERIA CAN BE GROWN” — 3 OF 4 POINTS

On agar or in a broth / in nutrients — “grown in a Petri dish” is ignored · **aseptic technique** · incubated at a **stated temperature in the range 25–35 °C** · one more culture detail (pH, growth factors, 24 h to 1 week).

“In a Petri dish at a suitable temperature” scores **zero** of those two marks. Name the medium; state a number.

Aseptic technique: autoclave at 121 °C for 15 min; disinfect the surface; work **close to a Bunsen flame**; flame the loop before and after; flame bottle necks; lift the lid **slightly and at an angle**; tape but **do not seal** — sealing creates **anaerobic** conditions favouring harmful anaerobes; incubate **inverted at 25 °C or below** in schools; autoclave before disposal.

Measuring growth	Advantage / disadvantage
Cell count	Quick, absolute; counts living and dead
Dilution plating	Counts only living ; assumes 1 colony = 1 cell; 24-48 h
Dry mass	Direct measure; destructive and slow
Turbidity	Quick, continuous; counts dead cells and debris; needs a calibration curve

Cells per $\text{cm}^3 = (\text{colonies} \div \text{volume plated}) \times \text{dilution factor}$. Use a plate with **30-300 colonies** — fewer is unreliable, more and colonies merge.

Why one calibration curve will not serve all species: different **cell sizes, shapes and pigmentation** scatter and absorb light differently, so the same optical density means a different cell number.

THE GROWTH CURVE

Lag — synthesising enzymes, taking up water and nutrients.

Log — nutrients and space plentiful, binary fission at maximum rate, constant doubling time. **Stationary** — division equals death; nutrients limiting, toxic waste, lack of oxygen and space. **Death** — death exceeds division.

$k = (\log N_t - \log N_0) \div (0.301 t)$ — values **must be taken from the log phase**, which is a mark in itself. Also $N = N_0 \times 2^n$. **Why a log scale:** numbers span orders of magnitude, and the exponential phase becomes a **straight line whose gradient gives the growth rate**.

Bacteria	Viruses
Prokaryotic cell	Not a cell
Circular DNA + plasmids	DNA or RNA
Peptidoglycan wall	Capsid ; some have an envelope
70S ribosomes	None
Binary fission, independently	Only inside a host cell

Why viruses need living cells: they cannot replicate without a host; they need the host's **enzymes, monomers and ribosomes** — “to grow” is not accepted.

Why different viruses infect different tissues: each has a specific host cell and **attachment molecules or receptors** — “antigens” is ignored throughout this answer.

The four named viruses: **Ebola** (RNA, helical, enveloped), **TMV** (RNA, helical rod, plants), **HIV** (RNA plus reverse transcriptase, gp120 envelope, infects **T helper cells**), **λ phage** (DNA, head, tail and fibres). **Lytic:** attachment, entry, replication, assembly, lysis. **Latency:** viral nucleic acid integrated into host DNA and replicated with it.

IMMUNITY

THE MOST-PENALISED ERROR IN THE TOPIC

Macrophages present antigen **to T helper (CD4) cells** — **do not accept T killer cells, B cells or “other cells”**.

The specific response: macrophage engulfs the pathogen and presents antigen on MHC → presented to a **T helper** cell with a complementary receptor → T helper activated, divides by mitosis, releases **cytokines** → cytokines activate B cells and T killer cells → B cell divides into **plasma cells and B memory cells** → plasma cells secrete **antibodies** → T killer cells destroy infected cells with **perforin** → memory cells persist.

THE PRIMARY-RESPONSE DELAY — 4 MARKS, WITH A LINK-AGE RULE

The scheme states the answer **must be linked to a time delay or a many-staged, slow process** for full marks. Time for macrophages to bind, engulf and process; to present antigen to T helper cells; for T helper cells to activate B cells; for B cells to **differentiate into plasma cells**; for plasma cells to **synthesise antibody**; and there are **no memory cells** on first infection. A list of stages **without the time framing** is capped.

Phagocytosis: chemotaxis → recognises non-self via receptors → engulfs by endocytosis into a **phagosome** → **lysosome fuses** → hydrolytic enzymes digest → antigen presented.

Inflammation: mast cells release **histamine** → **vasodilation of arterioles** and increased capillary permeability → white blood cells and antibodies reach the site. “Capillaries dilate” is ignored.

Gut flora reduce pathogen growth because they **compete for nutrients or space** — “food” is ignored — and produce vitamin K.

Antibodies: Y-shaped, four chains, disulfide bonds, **two variable regions**, constant region, hinge. They act by **agglutination, neutralisation, opsonisation and complement activation**.

Pathogens evade immunity by **antigenic variability** (flu, HIV), hiding inside host cells (TB in macrophages), **preventing lysosome fusion**, latency, destroying immune cells (HIV) and suppressing the response.

	Active	Passive
Natural	Infection	Antibodies from the mother
Artificial	Vaccination	Injection of antibodies
Memory cells	Yes — long-lasting	No — immediate but short-lived

THE ANTIBIOTIC DEFINITION — 2 MARKS, THREE TRAPS

“A substance, chemical or drug” — **do not accept** “produced by a virus” — “that kills or inhibits the growth of bacteria” — **do not accept** “virus”; “pathogens” is ignored.

Bacteriostatic inhibits (the graph plateaus; tetracycline); **bactericidal** kills (the graph falls; penicillin). They target **peptidoglycan walls, 70S ribosomes, DNA gyrase and folate synthesis** — human cells have no wall and 80S ribosomes, and viruses have no wall, ribosomes or metabolism.

Herd immunity: vaccinated people will not develop the disease, so there are **fewer infected people to infect those who are not immune**, protecting those who cannot be vaccinated.

FORENSICS AND DECOMPOSITION

Method	Basis and limitation
Body temperature	Cools towards ambient, ~1.5–2 °C per hour, sigmoid. Useless once ambient is reached. Depends on body size, fat, clothing, wind, immersion. A cooler environment gives a shorter estimate
Rigor mortis	No ATP → cross-bridges cannot detach. Small muscles first at 4–6 h, complete ~12 h, passes off 24–36 h. Stiffness is subjective and the same stiffness occurs twice
Decomposition	Autolysis then putrefaction; depends on temperature, oxygen, water and location
Forensic entomology	Larval instars of known size take known times at known temperatures. Gives only a minimum time since death ; the body may have been moved or insects excluded

Insect succession: species appear **in a specific sequence** — *Lucilia* first, then *Chrysomya* as *Lucilia* is outcompeted, then *Ophyra*. The evaluation answer: any single method gives only a range, so use several together and give a range.

Saprotrophic digestion: enzymes **packaged by the Golgi into vesicles and released by exocytosis** → hydrolysis **outside** the organism → soluble products absorbed → respired, releasing CO₂. Fungal **hyphae give a large surface area**; detritivores increase surface area for decomposers.

PCR — AND THREE WAYS TO LOSE THE REAGENT MARK

95 °C denaturation (hydrogen bonds break) → **50–65 °C** annealing (primers bind) → **72 °C** extension (**Taq** DNA polymerase).

Reagents: primers; **mononucleotides** — “bases” is ignored, **“RNA nucleotides” is not accepted; Taq or DNA polymerase — “RNA polymerase” is not accepted**; buffer. **Why Taq**: from a thermophile, so not denatured at 95 °C. Copies = 2ⁿ × starting number.

Electrophoresis: cut with **restriction endonucleases** → load into agarose wells → DNA is **negatively charged because of its phosphate groups**, so it moves to the **anode** → **shorter fragments move further** → stain or probe and size against a ladder. Profiling uses **non-coding repeated sequences** because there is **no selection pressure keeping them constant**.

THE REJECTIONS BANK

Do not write	Write instead
“Immune” to a chemical or antibiotic	Resistant
“The selection pressure caused the mutation”	Mutation is random and occurs first
Antigen presented to T killer or B cells	T helper (CD4) cells
“Grown in a Petri dish at a suitable temperature”	On agar or in broth , at a stated 25–35 °C
Antibiotics “produced by a virus” or acting on viruses	Produced by microorganisms; act on bacteria
“Capillaries dilate”	Vasodilation ; blood flow increases
Gut flora “compete for food”	Compete for nutrients or space
Ecosystem without the abiotic environment	Organisms and the abiotic environment
Q ₁₀ with units; “global temperatures”	No units; atmosphere, surface or water
PCR with “bases” or “RNA polymerase”	Mononucleotides; Taq DNA polymerase
Viruses need host cells “to grow”	They need the host’s enzymes, monomers and ribosomes

WALKING INTO THE EXAM

Circle the command word and the mark allocation. On every asterisked question alternate description and explanation, and use connectives — two of the marks are for sustained reasoning. Quote figures with units, and use the examiner’s exact word.